

The background is a blurred image of a computer monitor. It shows a dark-themed code editor with various lines of code in different colors (blue, green, yellow). At the bottom, a taskbar is visible with several application icons, including what looks like a web browser, a file explorer, and some development tools. The overall image is out of focus, emphasizing the text overlay.

C#

Level 1

What Is *C#* And Why To
Learn



C#

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What Is C# And Why To Learn

Definition

C#

 C# (C Sharp) is a modern, general-purpose, strongly typed, object-oriented programming language developed by Microsoft.

 It is commonly used with the .NET platform.

Explanation

C# and .NET

-  C# is the programming language.
-  .NET is the platform providing runtime, libraries, and tools.
-  C# can be used to build many different types of applications.

Main Characteristics

-  Object-oriented.
-  Strongly typed.
-  High-level.
-  Modern and continuously evolving.
-  Integrated with the .NET ecosystem.

Procedures

1. Write C# source code.
2. Compile it into IL.
3. .NET runtime loads the application.
4. JIT can convert IL into native code.
5. The application executes on the target platform.

💡 Example

```
class Student
{
    public string Name;

    public void Study()
    {
        Console.WriteLine("Student is studying.");
    }
}
```

➡ Result

C# allows developers to create structured, object-oriented applications using the .NET platform.

Note

- ◆ *C#* is influenced by *C*, *C++*, and *Java*.
- ◆ Modern *C#* includes features such as generics, LINQ, async/await, pattern matching, records, and nullable reference types.
- ◆ *C#* can be used for desktop, web, cloud, mobile, gaming, IoT, and AI-related applications.

⊘ Warning

- ◆ Do not say "C# is a platform."
- ◆ C# is the language; .NET is the platform.
- ◆ C# is not simply "C++ with Garbage Collection."



Comparison



C#

- ✓ Programming language.
- ✓ Used to write application code.
- ✓ Strongly typed and object-oriented.
- ✓ Commonly used with .NET.



.NET

- ✓ Development platform.
- ✓ Provides runtime and libraries.
- ✓ Provides tools for building and running applications.
- ✓ Supports multiple programming languages.

✦✦ Summary

- ✓ C# is a modern programming language.
- ✓ It is strongly typed and object-oriented.
- ✓ It is commonly used with .NET.
- ✓ It can be used for web, desktop, cloud, mobile, gaming, and other applications.
- ✓ C# = Language.
- ✓ .NET = Platform.

Thank you 😊